HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **GPU computing applications** using **OpenCL** for parallel data processing, achieving a **30% reduction** in computation time for image analysis tasks.
- Developed and debugged **real-time graphics applications** utilizing **Vulkan**, focusing on optimizing rendering pipelines to improve frame rates by **20%** in complex scene simulations.
- Authored technical documentation for software projects, including API specifications and user guides for embedded systems, ensuring clarity and ease of adoption for downstream teams.
- Collaborated with systems and hardware engineers to define software requirements and interfaces for new product development, contributing to the successful integration of **embedded software** components.
- Participated in code reviews and performed regression testing for small feature additions to cloud edge software, identifying and resolving minor bugs to maintain code quality.
- Applied **C++ programming knowledge** to modify and validate existing utility programs, ensuring they met performance requirements and integrated seamlessly with system-level software.
- Containerized applications using **Docker** and configured basic resource limits for deployment on cloud infrastructure, supporting a consistent development and testing environment.

SKILLS

Programming Languages – C++, C#, Rust, Go, Python, Java

Graphics APIs & Frameworks – Vulkan, DirectX, Metal, OpenGL, WebGL, Unity, Unreal Engine

GPU Computing – OpenCL, CUDA, SYCL

Tools & Debugging – Nsight, PIX, GPA, RenderDoc, Git, Jenkins, Jira

Software Development – Embedded Software, Cloud Edge Software, Software Design, Unit Testing, Code Review, Agile Methodologies

Concepts – Image and video processing, Data Parallelism, Machine Learning Algorithms, Real-time Graphics, GPU Computing, Graphics Profiling and Tuning

PROJECTS

GPU Accelerated Image Processing Pipeline

- Developed a **GPU-accelerated image processing pipeline** using **CUDA** and **C++** to enhance filtering and analysis capabilities for large datasets, reducing processing time by **40%**.
- Implemented error handling and logging mechanisms within the pipeline to ensure robustness and facilitate debugging of complex computations on the GPU.
- Integrated the pipeline with a REST API built using **FastAPI**, enabling seamless access for other applications requiring high-performance image manipulation.

Real-time 3D Rendering Engine Component

- Contributed to a **real-time 3D rendering engine** by developing a new shader management module using **Vulkan** and **GLSL**.
- Optimized rendering calls and resource loading strategies, improving the average frame rate by **25%** in benchmark test scenes with complex geometry and textures.
- Collaborated with a small team to integrate the module, participating in design discussions and code reviews to ensure compatibility and maintainability.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)